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Moh Sabbir Saadat

Post-doctoral Scholar
Memorial Sloan Kettering Cancer Center

Highlights

I am a researcher specializing in ubiquitous wireless sensing and digital health perception using multi-modal data. My expertise spans digital signal processing, deep learning, and machine learning. My work focuses on building robust, data-driven systems that bridge advances in AI/ML with intelligent perception applications. My current project is part of a collaboration with an external team of researchers in the health domain. **I recently defended my Ph.D., and has accepted a post-doctoral research role at Memorial Sloan Kettering Cancer Center.**

Experience

Post-doctoral Scholar — **August, 2026 — Present**
Department of Medical Physics - Memorial Sloan Kettering Cancer Center — *Manhattan, NY, USA*

- Develop AI-based decision-support tools to inform clinical workflow, combining patient-state and operational data.
- Conduct translational research on multimodal sensing and clinical data integration methods to make clinical assessment.

Graduate Research Assistant — **January, 2019 — July, 2026**
SyReX Lab - University of South Carolina — *Columbia, SC, USA*

- Pursue novel ideas in ubiquitous wireless sensing and digital multi-modal health perception. *(9 publications, 1 patent).*
- Build synchronized multimodal systems *(radio, camera, audio, smartwatch)*, execute data preprocessing, and develop evaluation benchmarks.
- Build complex ML models, and execute large-scale distributed training on GPU backend to benchmark them for various tasks.
- Spearhead inter-disciplinary research collaboration through knowledge-sharing and visualization of outcomes.
- Assisted a 400+ level class on computer networks with 100+ students for 6 semesters (Socket programming with Java, Python, and C).
- Assisted incoming graduate students through knowledge-sharing on research ideas, research culture, and work logistics.

Executive Engineer — **October, 2016 — November, 2018**
Siemens Healthcare Limited — *Dhaka, Bangladesh*

- Oversaw the technical requirements of potential clients in the medical imaging domain.
- Built liaison between the engineering department and the existing clientele.

Education

Ph.D. in Computer Engineering, *University of South Carolina* — **July, 2026**
M.Sc. in Computer Engineering, *University of South Carolina* — **August, 2024**
B.Sc. in Electrical & Electronics Engineering, *Bangladesh University of Engineering & Technology* — **March, 2016**

Skills

Software Engineering: Python, MATLAB, C++, Java, HTML, Git version control, Shell scripting, Linux.
ML libraries: PyTorch, TensorFlow, Keras, TensorBoard, Ray Tune, Scikit-learn, MATLAB Deep Learning.
ML models: Graph Neural Network, Vision Transformers, Generative Adversarial Network (GAN), Autoencoder, Few-shot Learning, LSTM.
Foundation AI models: MobileNet, ResNet, Mediapipe, HRNet, Whisper, WhisperX, BERT, Sentence-BERT.
Computational skills: Range-angle, Micro-doppler, Antenna array (FMCW radar), PCD processing, Clustering, STFT, Wavelet, DTW.
Tools: Latex, Gnuplot, Inkscape, Onshape CAD, TI mmWave Studio, ROS, Codex, Github Copilot.

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Projects

- **Vision and Audio-based digital health perception for post-stroke recovery assessment** *(Multi-modal sensors, pose, ASR, time-series)*
  - Built multi-sensors prototype with fine hardware-software synchronization. *(4k camera, microphone array, smart-watch, wireless)*
  - Facilitating the collection of clinical trials from stroke survivors and healthy subjects for post-stroke recovery assessment.
  - Translating conventional clinical practices to a vision- and audio-based AI-enabled automated system.
    * Assessing motor functionality from full-body images and facial abnormality from facial images. *(HRNet, Mediapipe)*
    * Image processing pipeline to mask identity and lighting conditions in facial images.
    * Audio noise cancellation and Audio Speech Recognition (ASR). *(Spectral noise cancellation, Whisper, WhisperX)*
    * Cognitive assessments through Large Language Models (LLM). *(BERT, Sentence-BERT, BLIP)*
    * Exploring speech intelligibility through phonetic comparison.

- **Co-existence of human-activity sensing on indoor wireless system** *(3D pose, Range-angle, μ-doppler, Graph Neural Network, Xformer)*
  - Graph neural network pipeline to overcome low-rate sensing signal due to co-existing networking.
  - Graph and Recurrent neural network (LSTM) to estimate sequences of 3D keypoints coordinates.
  - Deployment of activity recognition model on a real-time system ($\sim$98% prediction accuracy with $\sim$2s initial latency).
  - Explored the Transformer model to develop an end-to-end system leveraging range-angle-doppler heatmaps sequence.

- **Imaging hidden objects with handheld millimeter-wave devices** *(SAR, Compressed sensing, cGAN, Ray-tracing)*
  - Leverage Synthetic Aperture Radar (SAR) principle to enable imaging from a small antenna array.
  - Overcome sparse sampling and motion non-linearity with signal processing methods. *(compressed sensing, unsupervised clustering)*
  - Overcome information loss in reflected wireless signal through cGAN-based image super-resolution.
  - Overcome the lack of ground truth data through ray-tracing based simulation of wireless signal reflections.

Recent Publications

- *[Under review]* **Moh Sabbir Saadat**, Ryan Titus, Haley Verkuilen, Phil Fleming, Souvik Sen, Sanjib Sur. "*NeuroMotion:* A Contactless and Automated Approach to Assessing Post-Stroke Severity in Facial and Motor Functions." *IEEE Transactions on Neural Systems and Rehabilitation Engineering* **[TNSRE 2026]**

- *[To be submitted]* **Moh Sabbir Saadat**, Ryan Titus, Haley Verkuilen, Phil Fleming, Souvik Sen, Sanjib Sur. "*NeuroConv:* An Automated Approach to Assessing Post-Stroke Speech Functions from Audio." *IEEE Transactions on Neural Systems and Rehabilitation Engineering* **[TNSRE 2026]**

- **Moh Sabbir Saadat**, Ryan Titus, Haley Verkuilen, Phil Fleming, Sanjib Sur, Souvik Sen. "A Contactless and Automated Approach to the Acute Stroke Assessment." *AHA International Stroke Conference (February 2025)* **[AHA ISC 2025]**

- **Moh Sabbir Saadat**, Sanjib Sur. "Enabling Coexistence of Indoor Millimeter-Wave Networking and Human Activity Sensing." *2024 IEEE/ACM International Conference on Connected Health: Applications, Systems, and Technologies (June 2024)* **[CHASE 2024]**

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- **Moh Sabbir Saadat**, Sanjib Sur. "Human Activity Sensing from Low-rate Samples under Integrated Networking." *2024 IEEE/ACM International Conference on Connected Health: Applications, Systems, and Technologies (June 2024)* **[CHASE 2024]**
- **Moh Sabbir Saadat**, Sanjib Sur, Srihari Nelakuditi. "Aquilo: Temperature-Aware Scheduler for Millimeter-Wave Devices and Networks." *The Elsevier High Confidence Computing journal* **[Elsevier HCC 2024]**
- Edward Sitar, **Moh Sabbir Saadat**, Sanjib Sur. "A Millimeter-Wave Wireless Sensing Approach for At-Home Exercise Recognition." *Proceedings of the ACM International Conference on Mobile Systems, Applications, and Services (June 2022)* **[MobiSys 2022]**
- Hem Regmi, **Moh Sabbir Saadat**, Sanjib Sur, Srihari Nelakuditi. "SquiggleMilli: Approximating SAR Imaging on Mobile Millimeter-Wave Devices." *Proceedings of the ACM on Interactive, Mobile, Wearable, and Ubiquitious Technologies (September 2021)* **[IMWUT 2021]**
- Hem Regmi, **Moh Sabbir Saadat**, Sanjib Sur, Srihari Nelakuditi. "ZigZagCam: Pushing the Limits of Hand-held Millimeter-Wave Imaging." *ACM International Workshop on Mobile Computing Systems and Applications (February 2021)* **[HotMobile 2021] (Best Poster Runner-up Award)**
- **Moh Sabbir Saadat**, Sanjib Sur, Srihari Nelakuditi. "A Case for Temperature-Aware Scheduler for Millimeter-Wave Devices and Networks." *The 28th IEEE International Conference on Network Protocols (October 2020)* **[ICNP 2020]**
- **Moh Sabbir Saadat**, Sanjib Sur, Srihari Nelakuditi. "Bringing Temperature-Awareness to Millimeter-Wave Networks." *ACM International Conference Mobile Computing and Networking (September 2020)* **[MobiCom 2020]**
- **Moh Sabbir Saadat**, Sanjib Sur, Srihari Nelakuditi, Parmesh Ramanathan. "MilliCam: Hand-held Millimeter-Wave Imaging." *The 29th IEEE Conference on Computer Communications and Networks (August 2020)* **[ICCCN 2020]**

Patent

Sanjib Sur, Moh Sabbir Saadat, Srihari Nelakuditi, *"Heat Dissipation for Millimeter-wave Devices with Antenna Switching"* (Granted: February 2023)

Awards

- **Student travel grant, CHASE'24**: *Presenting work on enabling sensing on integrated networking system*
- **Best poster runner-up at ACM HotMobile'21**: *For early work on hand-held millimeter-wave imaging*
- **Saluting the Nation Builders of Tomorrow 2008/2010, The Daily Star** : *For outstanding achievement in GCE O/A level*

Professional services

- **External Reviewer**
  → IEEE Transactions on Mobile Computing 2022
  → IEEE/ACM Transactions on Networking 2023
  → IEEE International Conference on Mobility, Sensing, and Networking (MSN'23)
  → IEEE/ACM International Conference on Internet of Things Design and Implementation (IoTDI'23)
  → EAI Mobiquitious 2022
  → Pervasive and Mobile Computing 2025
- **Other Services**
  → Served as the Vice President of Bangladesh Student Association at the University of South Carolina 2021-2022
  → Served as the General Secretary of Bangladesh Student Association at the University of South Carolina 2020-2021
  → Served as an adjudicator at the NALSAR University Debating Tournament, Hyderabad, India 2012